# DEBUT Challenge – Project Narrative

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BIOE 141A/B, 2025–2026

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## 1. Abstract

Incidental pulmonary nodules are frequently detected in low-risk patients, yet current workup guidelines—developed for high-risk smokers—lead to repeated imaging, invasive biopsies, and surgical resection of nodules that are ultimately benign. Nearly 40% of bronchoscopic biopsies yield benign tissue [**avendano_outcomes_2024**], and 12.8% of surgical resections remove benign nodules [**archer_surgical_2023**], imposing unnecessary risk and cost. We developed a fluorescence-based split-aptamer biochip [**wang_microrna_2022**] that measures three circulating microRNAs (miR-21-5p, miR-155-5p, miR-1268b-5p) associated with lung adenocarcinoma to non-invasively triage patients with incidental nodule findings. Two short DNA strands assemble into a ligand-binding pocket only when the correct target miRNA is present, stabilizing the fluorophore Auramine O and generating a turn-on signal. We computationally designed split-aptamer sequences using AlphaFold3 and AutoDock Vina, then experimentally confirmed target-specific fluorescence activation in buffer, biological media, and cancer cell-conditioned media. A machine learning classifier trained on 3,743 public serum miRNA profiles (GEO GSE137140) achieved a holdout AUC of 0.987, demonstrating sufficient predictive information for malignant-versus-benign classification. We prototyped a four-chamber 3D-printed biochip with gravity-driven flow splitting and a low-cost ($54) LED-and-filter fluorescence reader. This platform enables rapid, non-invasive lung nodule risk stratification that could reduce unnecessary biopsies, lower healthcare costs, and improve patient outcomes.

## 2. Description of the Problem and Need

We would like to indicate our interest in the Technologies for Cancer Prevention, Diagnosis or Treatment Prize ($15,000), as our project directly addresses early-stage lung cancer nodule classification.

Incidental pulmonary nodules are common findings in CT scans, even in patients without traditional risk factors [**schmid-bindert_incidental_2022**]. A study of over 10,000 non-smokers found that at least 42% have a lung nodule [**cai_distribution_2024**], yet only 0.3% of never- or light-smokers actually had pathologic lung cancer [**du_airflow_2022**], implying most incidental nodules are benign. However, current nodule management guidelines were developed based on high-risk smokers and tend to err on the side of caution. This leads to many low-risk patients undergoing repeated follow-up scans, invasive biopsies, or even surgery for nodules that ultimately turn out benign [**cai_distribution_2024**].

The clinical consequences are substantial. Of the $38.3 million spent by Medicare on lung cancer diagnosis, 43.1% of costs stemmed from patients receiving unnecessary biopsies for nodules that were not cancerous [**chiu_costs_2021**]. The risk of complications associated with even the least invasive biopsy (percutaneous needle biopsy) can be as high as 15.0% pneumothorax [**zhang_biopsy_2020**], and 19.3% of patients experience an adverse event after receiving a lung biopsy [**lewis_reducing_2025**]. Moreover, 12.8% of surgical resections remove nodules that are ultimately benign [**archer_surgical_2023**], exposing patients to the risks of major surgery without therapeutic benefit.

Early diagnosis matters profoundly: patients diagnosed with Stage IA lung cancer have a >75%

five-year survival rate, versus 5% at late stages, yet only 15% of patients are diagnosed early because the disease is largely asymptomatic [**afridi_minimally_2025**]. A non-invasive method to distinguish benign from malignant nodules at the point of incidental finding would reduce unnecessary procedures, lower costs, and enable faster treatment access.

The primary patient population consists of adults with pulmonary nodules detected incidentally during imaging for unrelated conditions, most commonly between ages 45–70 [**chen_prevalence_2024**]. Current non-imaging approaches (blood, sputum, breath-based tests) often direct patients back to CT imaging rather than providing independent nodule risk assessment [**mazzone_clinical_2024**].

Our need statement: A way to distinguish benign from malignant lung nodules in low-risk patients with incidental findings in order to reduce unnecessary biopsies and procedures.

## 3. Project Objective Statement

We are developing a fluorescence-based split-aptamer biochip diagnostic that measures a panel of three circulating microRNAs to triage patients with incidental lung nodule findings for biopsy with high accuracy.

Our system leverages split-aptamer molecular recognition: for each miRNA target, two short DNA strands (St1, St2) are designed such that the presence of the correct target miRNA promotes their assembly into a three-way junction that stabilizes the fluorophore Auramine O, generating a measurable fluorescence turn-on signal. The biochip routes a small-volume patient sample (serum) into three parallel sensing chambers - one per miRNA target - plus an internal negative control. Fluorescence readout from each chamber is normalized and fed into a trained machine learning classifier that outputs a malignancy risk score.

Unlike existing liquid biopsy approaches that require expensive sequencing or PCR amplification, our platform achieves direct, amplification-free miRNA detection through a novel split-aptamer architecture adapted from Wang et al. [**wang_microrna_2022**]. We computationally retargeted validated aptamer scaffolds to new miRNA biomarkers using AlphaFold3-guided sequence design: a semi-rational approach that combines structural biology with aptamer engineering. The integration of molecular sensing, microfluidic sample handling, and on-device machine learning classification into a single low-cost ($54 reader, low-single-digit-dollar disposable chip) platform distinguishes our approach from conventional laboratory-based molecular diagnostics.

Through a combination of computational ablation analysis and wet-lab validation, we converged on a final three-miRNA panel: miR-21-5p and miR-155-5p (oncogenic miRNAs overexpressed in lung adenocarcinoma, associated with suppression of apoptotic markers [**wang_microrna_2022**]) and miR-1268b-5p (a miRNA whose underexpression is highly correlated with lung cancer in profiling studies [**zhu_mir-1268b_2017**]). miR-126-5p was dropped from the original four-marker panel based on evidence from both our computational and wet-lab work; computational ablation of miR-126-5p showed only 0.001 AUC loss upon removal, and our wet-lab experiments detected upregulation of the miRNA in the cancer condition, contrary to its reported tumor-suppressive role [**chen_microrna-126_2021**].

Key specifications.

- Target accuracy: >77% (exceeding quantitative radiomic models for small solid nodule classification [**mao_quantitative_2019**]); achieved holdout AUC of 0.987 on serum miRNA data
- Cost: <$600 per test (below median out-of-pocket biopsy cost [**zheng_patient_2021**]); reader hardware $54, disposable chip low single-digit dollars

- Turnaround time: <7 days to receive results (current benchtop assay: ∼1 hour reaction + readout)
- Invasiveness: standard blood draw only; no anesthesia required
- Operator skill: designed for trained technician or general practitioner operation

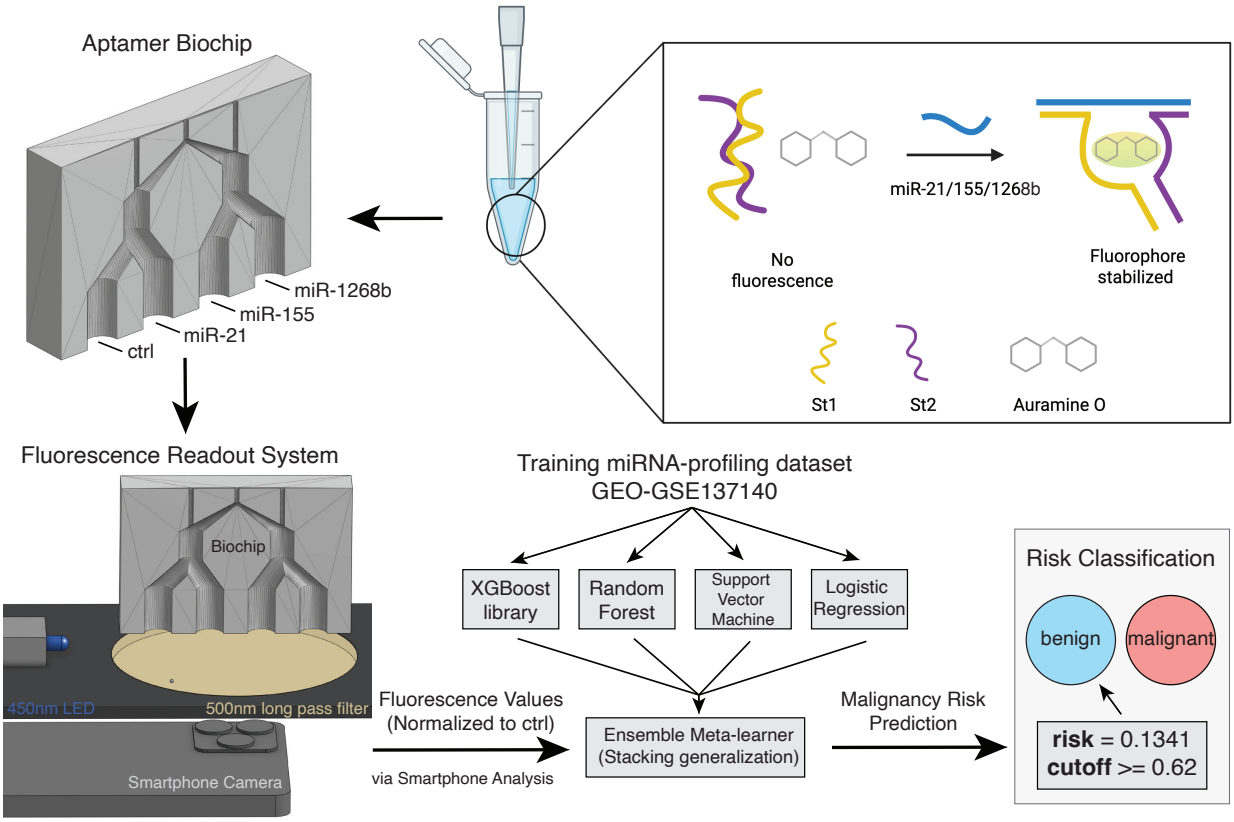


**Figure 1: System overview.** A small-volume serum sample is routed into three parallel sensing chambers plus a negative control, each containing a miRNA-specific split-aptamer pair and Auramine O. Target-dependent assembly stabilizes AO and produces a fluorescence turn-on signal, which is measured by a fluorescence readout system consisting of Auramine O-specific excitation LED and long-pass emission filter at a 90º angle to miRNA samples. Normalized fluorescence values are fed into an XGBoost classifier (trained on public serum miRNA data, AUC = 0.987) to output a malignancy risk score.

## 4. Documentation of the Design

### 4.1 Split-Aptamer Molecular Recognition

Our sensing platform adapts the validated split-aptamer scaffolds reported by Wang et al. [**wang_microrna_2022**], in which two DNA strands (St1, St2) assemble into a functional ligand-binding pocket only in the presence of a target miRNA, enabling fluorophore stabilization and fluorescence activation. To retarget these constructs for our biomarker panel, we preserved the structural domains that govern folding and ligand binding (LigBR sequence, reported bulge sizes, and scaffold-specific P3 lengths) and modified only the hybridizing regions (P2 on St1 [9 nt] and P4 on St2 [10 nt]) with sequences complementary to each target miRNA [**wang_microrna_2022**]. For each miRNA target, we designed three candidate aptamer pairs using the a2, a6, and a7 scaffold variants, which differ in bulge size and provide structural diversity for experimental screening.

### 4.2 Computational Validation

To evaluate whether retargeted constructs preserve target-dependent assembly, we used AlphaFold 3 to predict the structure of each St1–St2–miRNA triplet. For each candidate, we input three entities (St1 DNA, St2 DNA, target miRNA RNA) and assessed predicted three-way junction formation and overall model confidence (ipTM score). We also ran the original miR-21 a2/a6/a7 constructs as baselines. AlphaFold3 predicted target-dependent three-way junction formation across all miRNA–aptamer triplets, and without a miRNA present, the predicted structures showed far lower confidence, confirming that the miRNA is necessary for proper assembly.

We assessed fluorophore-pocket feasibility using AutoDock Vina [**eberhardt_autodock_2021**, **trott_autodock_2010**] by docking Auramine O into the predicted complexes. The top docking pose for miR-21 achieved an affinity of $-7.10$ kcal/mol, with the AO fluorophore positioned within the predicted LigBR binding pocket and showing hydrogen bonding interactions with St1 base pairs. Docking simulations across all 12 aptamer–miRNA combinations identified accessible binding pockets consistent with the fluorescence turn-on mechanism.

### 4.3 Machine Learning Classification

To validate that the selected miRNA biomarkers contain sufficient predictive information, we analyzed a publicly available circulating miRNA expression dataset (GEO GSE137140 [**wang_microrna_2022**]) containing 3,743 serum samples (1,566 cancer, 2,177 control). We conducted a full-factorial ablation study over all 15 non-empty subsets of the four candidate miRNAs and evaluated nine machine learning classifiers.

miR-1268b-5p emerged as the single most predictive biomarker, achieving an AUC of 0.977 as the sole miRNA feature. Dropping miR-126-5p resulted in only 0.001 AUC decrease, supporting its removal from the panel. The final three-miRNA panel (miR-21-5p, miR-155-5p, miR-1268b-5p) plus demographic covariates (age, sex [**blechter_age_2024**]) was evaluated across all nine classifiers; XGBoost achieved the best holdout AUC of 0.987 with 95.2% sensitivity and 94.5% specificity. SHAP feature importance analysis confirmed miR-1268b-5p as the dominant predictor (mean $|\text{SHAP}| = 2.92$), followed by miR-155-5p (1.07) and miR-21-5p (0.76). SHAP interaction analysis confirmed that demographic variables do not confound the miRNA biomarker signal.

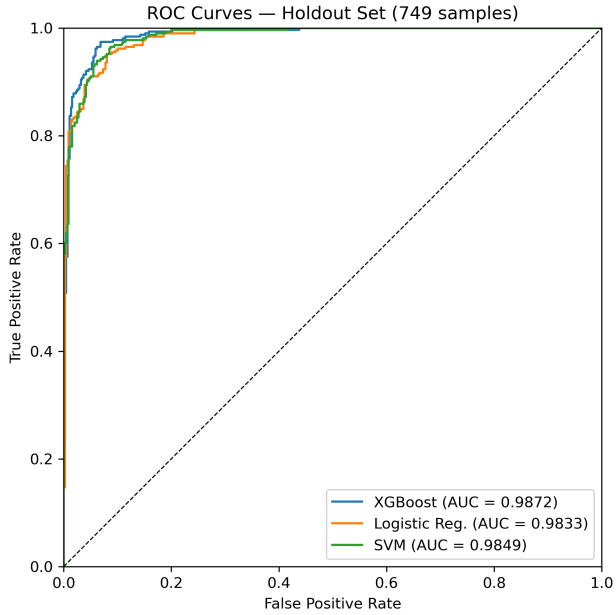


**Figure 2: Holdout set ROC curves.** ROC curves for three representative classifiers on the 749-sample holdout set. All models achieve AUC $> 0.98$, with XGBoost performing best (AUC $= 0.987$).

## 5. Prototype of the Final Design and Proof of Function

### 5.1 Fluorescence Validation in Buffer and Media

We experimentally validated the split-aptamer fluorescence response for miR-21, miR-126, and miR-155 in both fluorescence assay buffer (FAB) and biological media (DMEM). At equimolar concentrations (2 $\mu$M aptamer, 2 $\mu$M miRNA), we observed statistically significant fluorescence differences between the target miRNA condition and all controls (scrambled miRNA, no miRNA, fluorophore only) for each aptamer system in both buffer and media ($p < 0.001$ for miR-21 and miR-155). miR-155 showed the strongest signal across both experiments.

Limit-of-detection experiments using serial dilutions (0–100 nM) revealed an LOD of $\sim$25 nM for miR-155 ($R^2 = 0.98$) and $\sim$50 nM for miR-21 ($R^2 = 0.79$). Iterative redesign of the miR-126 aptamer scaffold (increasing the bulge region by 1 nucleotide) yielded a significantly improved variant (miR-126n), and a newly designed miR-1268b aptamer system was also validated in both buffer and media.

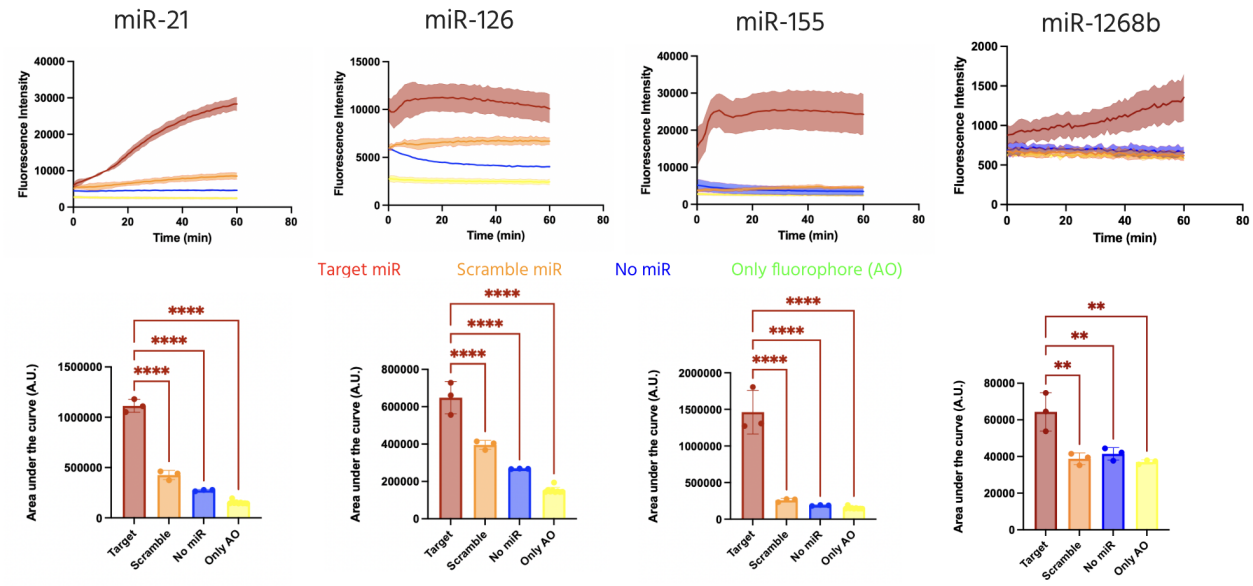


**Figure 3: Fluorescence validation in cell media.** (top) Raw fluorescence time courses for each miRNA target and controls (scramble, no miRNA, fluorophore only) in cell media (DMEM), measured over one hour ($n = 3$). (bottom) AUC analysis for each miRNA relative to its controls.

### 5.2 Cell Line Discrimination

Conditioned media from A549 lung adenocarcinoma and HBE135-E6E7 healthy lung epithelial cell lines were tested to determine whether the aptamer systems could detect differences between cancerous and non-cancerous conditions. Across all four targets, A549-conditioned media generally produced higher fluorescence signals than HBE-conditioned media, consistent with elevated oncogenic miRNA activity in cancer cells. While media-dependent background effects were identified as a confounding factor requiring further optimization, the results demonstrated that the split-aptamer sensors retain target responsiveness in biologically relevant matrices.

### 5.3 Biochip Prototype

We developed a physical biochip prototype through four iterative design cycles (V1–V3.2). The final V3.2 prototype features a vertical cartridge geometry with a single top inlet that splits into four symmetric channels feeding four bottom wells (three assay chambers plus one negative control). Each well is approximately 9 mm in diameter (matching 96-well plate format), and the chip follows a standard microscope-slide footprint (75 mm × 25 mm). The bottom of each well is sealed with a glass microscope slide for optical interrogation. Air outlet vents in each well enable gravity-driven filling.

We also assembled a low-cost fluorescence readout subsystem using a 450–460 nm blue LED with PWM-controlled luminance (via Arduino and LED driver), a long-pass emission filter, and smartphone camera-based imaging. The reader hardware costs approximately $54 per unit (including a proper 530 nm long-pass filter), while the disposable biochip cartridge remains in the low-single-digit-dollar range—orders of magnitude cheaper than conventional benchtop fluorescence instrumentation.

### 5.4 Summary of Results

**Table 1:** Summary of key experimental and computational results across all milestones.

| Milestone | Key Result |
|---|---|
| 1. Computational validation | AlphaFold3 confirmed three-way junction formation for all miRNA–aptamer triplets; AutoDock Vina docking affinity −7.10 kcal/mol for top miR-21 pose |
| 2. Buffer/media validation | Statistically significant target-specific fluorescence ($p < 0.001$); LOD 25 nM (miR-155), 50 nM (miR-21) |
| 3. Cell line testing | A549 cancer media showed higher fluorescence than HBE healthy media across targets |
| 4. ML classification | Holdout AUC 0.987 (XGBoost); three-miRNA panel validated; miR-1268b-5p identified as dominant predictor |
| 5. Hardware prototype | Four-chamber biochip (V3.2) with gravity-driven flow; $54 fluorescence reader prototype |

## 6. Video

*[VIDEO LINK PLACEHOLDER — We will record a video before we submit to DEBUT and insert a link to it here.]*

# Part 2: IP, Regulatory, and Commercialization Analysis

## 7. Patent Search and Prior Art Assessment

### 7.1 Prior Art Landscape

We conducted a patent search across three technology domains relevant to our device: (1) split-aptamer diagnostics, (2) miRNA-based lung cancer panels, and (3) microfluidic biochip platforms for nucleic acid detection.

The foundational split-aptamer fluorescence detection technology was described by Wang et al. [**wang_microrna_2022**] in the peer-reviewed literature but to our knowledge, does not appear to have been patented as a standalone diagnostic platform. Existing aptamer biosensor patents generally cover different transduction mechanisms (electrochemical, colorimetric, or nanoparticle-based) rather than the split-aptamer fluorescence turn-on approach we employ.

Multiple patents cover miRNA panels for lung cancer detection, generally claiming specific panel compositions different from our three-miRNA combination (miR-21-5p, miR-155-5p, miR-1268b-5p) and relying on PCR or sequencing-based quantification rather than direct aptamer-mediated detection. Our specific panel composition and detection modality appear to be distinct from existing claims.

The microfluidic biochip space is densely patented with broad claims covering channel geometries, sample routing, and integrated detection. However, the combination of split-aptamer sensing chemistry with a gravity-driven multi-chamber chip format and low-cost LED-based fluorescence readout appears to be novel in the patent landscape. The closest patent issued to our proposed solution is US20100240544A1, which consists of a biochip with immobilized aptamer "templates" on the surface that allow for RNA transcription to occur to generate the RNA aptamers in situ [**liu_aptamer_2010**]. However, this patent was not only abandoned since its publication in 2010 but furthermore contrasts from our solution in that we use DNA split-aptamers and require neither in-situ transcription nor hybridization of the aptamers onto the chip, which in theory reduces the cost of our solution.

### 7.2 Patentability Assessment

We believe the following aspects of our technology may be patentable:

1. The specific combination of computationally retargeted split-aptamer sequences for a defined miRNA panel (miR-21-5p, miR-155-5p, miR-1268b-5p) with Auramine O fluorescence readout
2. The integrated device architecture combining gravity-driven multi-chamber biochip with low-cost LED/filter fluorescence detection and on-device ML classification
3. The semi-rational AlphaFold3-guided approach to retargeting split-aptamer scaffolds for new nucleic acid targets

A freedom-to-operate analysis should be conducted before commercialization, particularly with respect to broad aptamer biosensor patents and the foundational Wang et al. [**wang_microrna_2022**] split-aptamer work.

## 8. Anticipated Regulatory Pathway

Our device would be classified as an in vitro diagnostic (IVD) device by the FDA. Given that it provides adjunctive risk stratification information (triage to biopsy) rather than a standalone cancer diagnosis, we anticipate pursuing one of two regulatory pathways:

If a suitable predicate device can be identified among existing FDA-cleared liquid biopsy or molecular diagnostic platforms, a 510(k) submission demonstrating substantial equivalence would be the fastest route to market. The key argument would be that our device provides a similar type of clinical information (molecular biomarker-based cancer risk assessment from a blood sample) using a different but functionally equivalent detection technology.

If no adequate predicate exists—particularly because our device uses a novel split-aptamer detection chemistry rather than PCR or sequencing—a De Novo classification request may be more appropriate. This pathway would establish a new regulatory classification for split-aptamer-based molecular diagnostics and define appropriate special controls (e.g., analytical performance requirements, clinical validation study design, quality system requirements).

In either case, clinical validation studies would be required to demonstrate analytical performance (sensitivity, specificity, limit of detection, reproducibility) and clinical performance (correlation with biopsy outcomes) in the intended-use population.

## 9. Reimbursement

For molecular diagnostic tests, reimbursement is primarily governed by the Clinical Laboratory Fee Schedule (CLFS) under Medicare. Our test would likely fall under existing CPT codes for molecular pathology procedures:

- CPT 81479 (Unlisted molecular pathology procedure): applicable for novel molecular tests not yet assigned a specific code. This would be the initial reimbursement pathway while seeking a unique CPT code.
- Proprietary Laboratory Analyses (PLA): once clinically validated, we would apply for a proprietary PLA code that specifically describes our miRNA panel and interpretation algorithm.

Existing liquid biopsy and molecular diagnostic tests for lung cancer are typically priced in the thousands of dollars per test. We would target a reimbursement rate of $300–$600, which is substantially lower than existing alternatives while remaining well above our projected cost of goods ($2–$5 per cartridge). The economic argument for payer coverage is strong: at the current rate, 43.1% of Medicare lung cancer diagnostic spending goes to biopsies on ultimately benign nodules [**chiu_costs_2021**]. A $300–$600 screening test that reduces the rate of unnecessary biopsies (each costing $600+ out-of-pocket [**zheng_patient_2021**], with total procedural costs much higher) would generate net savings for the healthcare system.

## 10. Estimated Manufacturing Costs

### 10.1 Fluorescence Reader (Reusable)

The reader is intended as a modest one-time equipment purchase for the clinical laboratory or point-of-care site. Our prototype cost breakdown demonstrates the feasibility of an extremely low-cost reader:

| Component | Unit Cost ($) |
|---|---|
| 450–455 nm LED | 2.00 |
| LED driver | 2.22 |
| Heat sink | 2.12 |
| Inline switch | 3.50 |
| 12 V power adapter | 8.99 |
| 530 nm long-pass filter | 35.00 |
| **Total** | **53.82** |

At manufacturing scale with injection-molded housing and integrated electronics, we estimate the reader could be produced for $100–$200 per unit and sold at $300–$500, which is orders of magnitude cheaper than conventional plate readers ($10,000–$50,000).

### 10.2 Disposable Biochip Cartridge (Per-Test)

| Component | Estimated Cost ($) |
|---|---|
| 3D-printed/injection-molded cartridge body | 0.50–1.00 |
| Glass microscope slide (optical window) | 0.10–0.25 |
| Pre-loaded split-aptamer reagents (3 targets + control) | 1.00–3.00 |
| Auramine O fluorophore | <0.10 |
| **Estimated per-test COGS** | **$2–$5** |

At a projected selling price of $50–$100 per disposable cartridge (well below the reimbursement target), the per-test margin supports a sustainable business model. The separation between reusable reader and disposable cartridge is favorable for deployment in resource-constrained settings.

## 11. Potential Market and Impact

### 11.1 Market Size

Lung cancer is the leading cause of cancer death worldwide. In the United States:

- Roughly 57,000 lung cancer resections are performed per year [**potter_assessing_2023**]
- 19% of women and 9% of men with lung cancer are non-smokers [**dubin_lung_2020**]
- Pulmonary nodules are detected in 27–30% of screening cohorts [**chen_prevalence_2024**], representing a large population requiring risk stratification

Our addressable market focuses on patients with incidental pulmonary nodule findings. For patients who receive surgery only, cancer-attributable costs are approximately $828 per month; when chemotherapy, radiation, or both are required, costs range from $4,242 to $8,287 per month [**sheehan_lung_2018**]. Of the $38.3 million spent by Medicare on lung cancer diagnosis, 43.1% went to biopsies on ultimately benign nodules [**chiu_costs_2021**]. Even capturing a fraction of this diagnostic workup volume at $300–$600 per test represents a significant commercial opportunity with clear cost savings for the healthcare system.

### 11.2 Pricing and Distribution

We envision a razor-and-blade model: the reader is sold at near-cost to clinical laboratories, pulmonology practices, and community health centers, with recurring revenue from disposable cartridge sales. Initial distribution would target:

1. Academic medical centers and lung cancer screening programs (early adopters)
2. Community pulmonology practices (where patients with incidental findings are most commonly managed)
3. Low-resource and global health settings (where the low cost of the platform provides particular value)

### 11.3 Clinical Impact

By enabling non-invasive molecular risk stratification at the point of incidental finding, our device has the potential to:

- Reduce the rate of unnecessary biopsies and surgical resections of benign nodules
- Lower complication rates associated with invasive diagnostic procedures (up to 15% pneumothorax risk per biopsy [**zhang_biopsy_2020**])
- Decrease healthcare costs—43.1% of Medicare lung cancer diagnostic spending currently goes to biopsies on benign nodules [**chiu_costs_2021**]
- Alleviate patient anxiety associated with prolonged surveillance and repeated imaging
- Improve access to molecular diagnostics in settings where sequencing-based liquid biopsy platforms are cost-prohibitive

The combination of low per-test cost, minimal infrastructure requirements, and rapid turnaround time positions our platform to democratize access to molecular lung cancer screening—particularly in community practices and global health contexts where existing solutions are impractical.